MIL-C-27336B(USAF)
20 September 1978
SUPERSEDING
MIL-C-27336A(USAF)
14 November 1963

MILITARY SPECIFICATION

CONVERTER, LIQUID OXYGEN TYPE GCU-17/A

This specification is approved for use by the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of liquid oxygen converter with a capacitance gaging element. The converter is designated Type GCU-17/A. The converter does not include warming coils. (See 6.1.1).

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-V-9050	Valve, Oxygen, Pressure Relief, Aircraft
MIL-C-25516	Connectors, Electrical , Miniature, Coaxial, Environment Resistant Type, General Specification for
MIL-C-25666	Converters, Liquid Oxygen, Capacitance Type Gaging, General Specification For
MIL-C-25674	Converter, Liquid Oxygen, Type ME-3

STANDARD

MILITARY

MIL-STD-831	Test Reports, Preparation of

Air Force - Navy Aeronautical

AN816	Nipple - Flared Tube and Pipe Thread

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: ASD/ENESS, Wright-Patterson AFB, OH 45433 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 1660

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DRAWING
Air Force
58J3535 Capacitance Chart - 25 Liter Liquid Oxygen Converter

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Qualification. Converters furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for opening of bids. (See 4.3 and 6.3).

3.2 General specification. The requirements of MIL-C-25666 shall apply as requirements of this specification with the exceptions and additions called out herein. When the two specifications conflict, this specification shall govern.

3.3 Design. The converter shall conform to the dimensions and general design as shown on figures 1 and 2. The converter shall have a minimum capacity of 25 liters of liquid oxygen measured at 1 atmosphere plus at least 3 percent of this volume to allow for the change in liquid density at the operating pressure. The converter shall be designed for an operating pressure of 300 psig. The converter shall be integrally shock mounted so that the entire converter assembly, including ports, is isolated from the mounting base. The natural resonant frequency of the vibration isolators shall fall between 8 and 30 cycles per second (cps). No resonance of the converter shall occur at a frequency less than 40 cps. The bosses for all pipe thread ports shall be fabricated from corrosion-resistant steel.

3.3.1 Schematic diagram. The converter shall be designed in accordance with the schematic diagram as shown on figure 3.

3.3.2 Design strength. The converter shall be designed to satisfy the dynamic force requirements specified in MIL-C-25666. When subjected to the acceleration test, the following exceptions to the values for G levels shall apply.

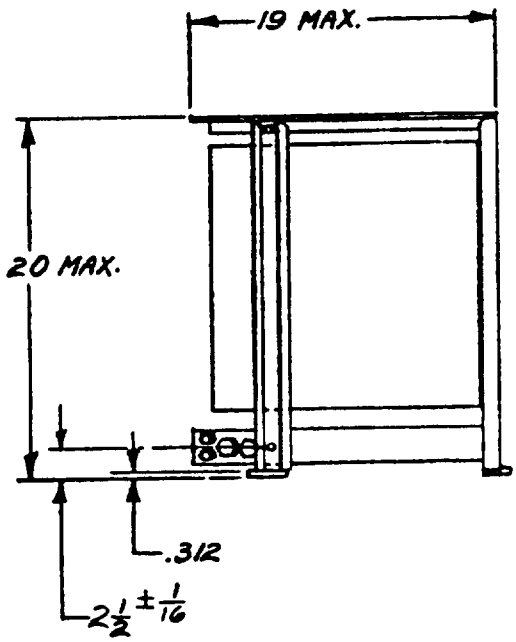
Vertical (down)	9.0G
Vertical (up)	3.0G
Lateral	4.0G

3.3.3 Relief valves. The relief valves shall conform to the performance requirements of types III and IV of MIL-V-9050 and shall be of such size and shape to fit the overall dimensions specified for the converter.

3.3.4 Weight. The weight of the complete converter shall not exceed 40 pounds.

3.3.5 Shock and shock safety. The requirements of 3.5.17 and 3.5.18 of MIL-C-25666 shall not apply as requirements of this specification.

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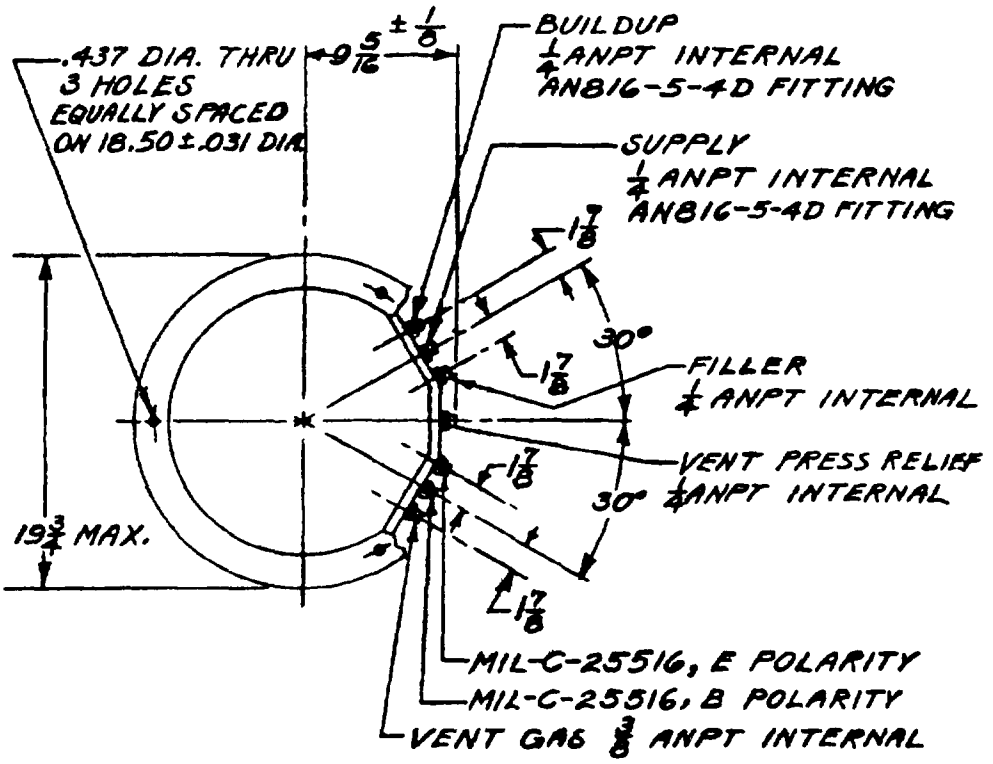


FULL LEG DESIGN OPTIONAL

UNLESS OTHERWISE SPECIFIED
DIMENSIONS ARE IN INCHES
TOLERANCES ON:
FRACTIONS = $\pm \frac{1}{64}$

FIGURE 1. Converter assembly.

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TYPE OF MOUNTING BASE OPTIONAL TO CONTRACTOR; EITHER THREE MOUNTING LEGS OR A MOUNTING RING, OR A TRIANGULAR FRAME.

DIMENSIONS IN INCHES
UNLESS OTHERWISE SPECIFIED
TOLERANCES:
FRACTIONS = ± 1/32
DECIMALS = ± .005

FIGURE 2. Converter design (Bottom view).

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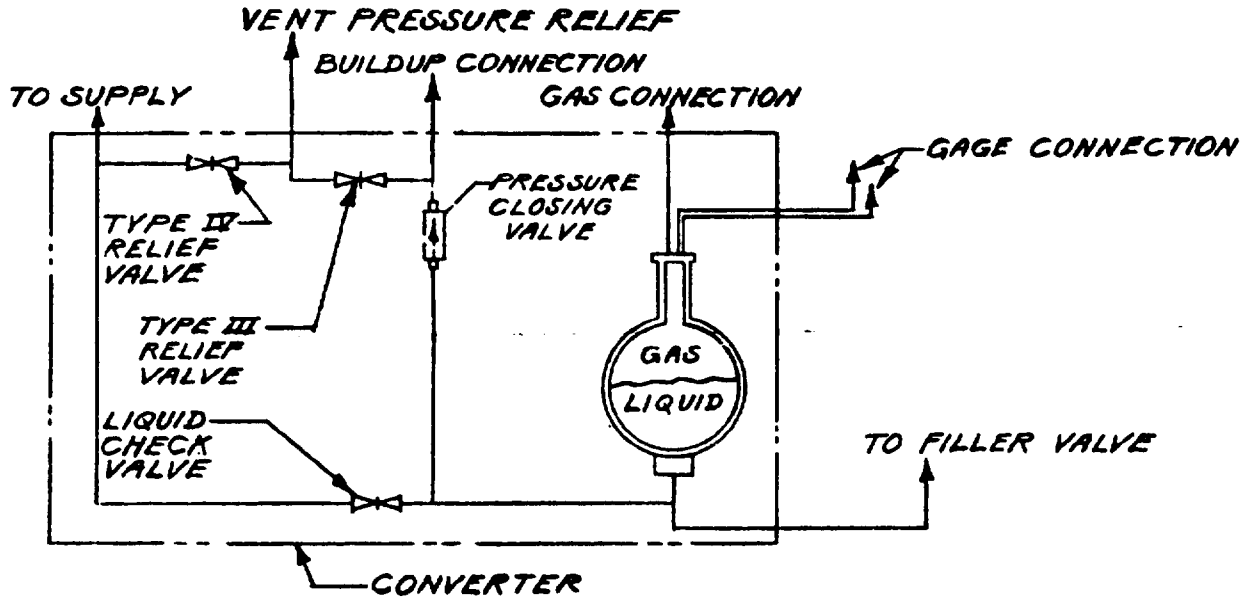


FIGURE 3. Schematic diagram of converter with capacitance gaging element.

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3.3.6 Overpressure safety device. Safety discs or other secondary overpressure devices, as specified in 3.6.6 of MIL-C-25666 shall not be required for this converter. The design of this converter incorporates two pressure relief valves, which satisfies 3.6.6 of MIL-C-25666.

3.4 Performance

3.4.1 Permanent volumetric expansion. The converter shall meet the performance requirements for Permanent Volumetric Expansion of MIL-C-25666 when subjected to a pneumatic or hydrostatic expansion pressure of 600 psig.

3.4.2 Evaporation loss. The evaporation loss of the converter shall not exceed 4 pounds of oxygen when the converter is tested as specified.

3.4.3 Final assembly pressure. The pressure used for the Final Assembly Pressure test shall be 320 psig.

3.4.4 Filling time. Filling time shall not exceed 10 minutes when the converter is tested as specified.

3.4.5 Buildup time. Buildup time shall not exceed 10 minutes when the converter is tested as specified.

3.4.6 Sensing element calibration. The capacitance sensing element shall have characteristics which provide for a linear relationship between electrical capacitance and volume of liquid oxygen sensed. The relationship shall be as shown on figure 4, using either the equations or the referenced drawing to determine tolerances. An example of use of the equations is as follows: At the 10 - liter test point, suppose the actual quantity of liquid oxygen is 10.1 liters. The measured capacitance must be between C maximum (max) and C minimum (min).

C min = 5.715 x 10.1 + 300.5 = 358.2 micromicrofarad (uuf)
C max = 5.875 x 10.1 + 306.5 = 365.8 micromicrofarad (uuf)

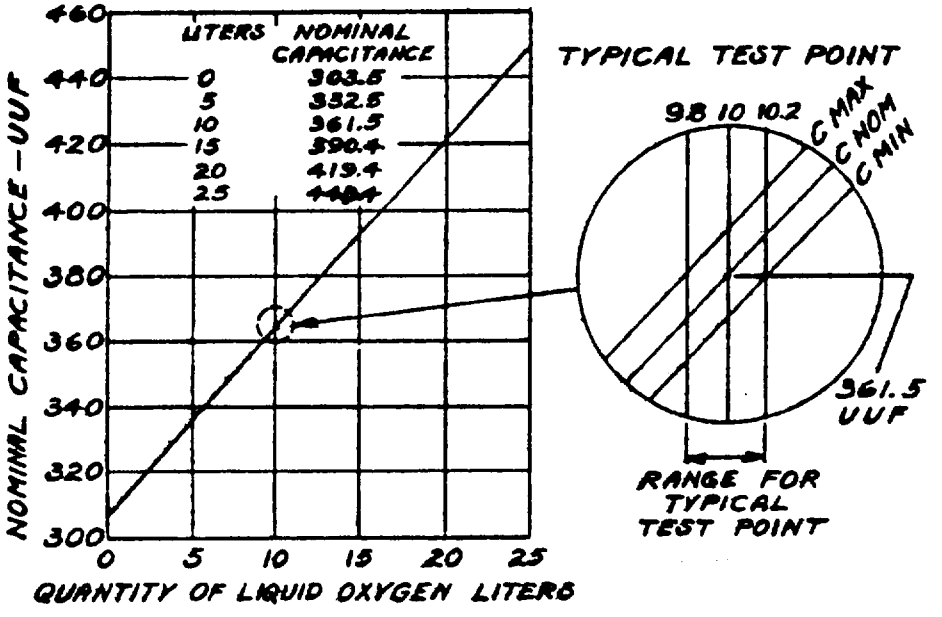
3.4.7 Delivery rate. The converter shall deliver constant gas flows up to 150 liters of gaseous oxygen per minute for not less than 2 hours with the operating pressure remaining between 295 and 315 psig after the first 15 minutes of operation. An intermittent increase or decrease of pressure by an additional 5 psig is allowable provided these variances do not exceed a 2-minute interval for any one pressure cycle.

3.4.8 Extreme temperature operation. The operating pressure shall remain between 295 and 320 psig after the first 15 minutes of flow at the low temperature and between 290 and 315 psig after the 15 minutes of flow at the high temperature. An intermittent increase or decrease of pressure by an additional 5 psig is allowable provided these variances do not exceed a 2-minute interval for any one pressure cycle.

3.4.9 Vibration. After the converter is vibrated in accordance with the procedure specified in MIL-C-25666 to a W_0 value of 0.15 G^2/Hertz, the evaporation loss shall not exceed 4.5 pounds of oxygen in the buildup condition.

3.4.10 Sensing element capacitance. The capacitance of the sensing element of the empty converter shall be 303.5 ±2.5 uuf.

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CALCULATION OF MAXIMUM AND MINIMUM CAPACITANCES

FOR VALUES LESS THAN 5 LITERS	FOR VALUES MORE THAN 5 LITERS
C MIN = 5.6L + 301.0	C MIN = 5.715L + 300.5
C MAX = 6.0L + 306.0	C MAX = 5.875L + 306.5

C = CAPACITANCE IN UUF
L = EXACT QUANTITY OF LIQUID OXYGEN AT TEST POINT IN LITERS.

CALIBRATION LIMITS, BASED ON THE ABOVE EQUATIONS, ARE SHOWN IN DIRECT READING GRAPH FORM ON DRAWING 58J3535.

FIGURE 4. Calculation of capacitances.

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3.5 Marking of ports. Ports shall be marked as shown on figure 2.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Sampling, inspecting, and testing procedures. Sampling, inspecting, and testing procedures shall be in accordance with MIL-C-25666.

4.3 Qualification testing

4.3.1 Qualification test samples. The contractor shall subject two or more converters to the qualification tests specified in 4.3.4.

4.3.2 Qualification test report. After the contractor completes the qualification tests, he shall prepare a qualification test report in accordance with MIL-STD-831 and shall furnish three complete copies of the report to the qualifying activity.

4.3.3 Qualification test sample for the qualifying activity. Along with the qualification test report, the contractor shall submit one converter. This sample may be a tested sample, except that it shall not be subjected to the test for gunfire specified in MIL-C-25666.

4.3.4 Qualification tests. Qualification tests shall consist of all tests described under test methods of MIL-C-25666.

5. PACKAGING

5.1 Preservation - packaging. The preservation - packaging of the converter shall be as specified in MIL-C-25666.

6. NOTES

6.1 Intended use. The converter covered by this specification is intended for use in B-52 airplanes.

6.1.1 Inactive for new design. This specification is inactive for new design. For new design MIL-C-25674 should be used.

6.2 Ordering data. Procurement documents should specify the following;

a. Title, number and date of this specification.

b. Selection of applicable levels of preservation-packaging and packing.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids,

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qualified for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Aeronautical Systems Division, ASD/ENECE, Wright-Patterson Air Force Base, Ohio 45433 and information pertaining to qualification of products may be obtained from that activity.

6.4 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodian:
Air Force - 11

Reviewer:
Air Force - 99

International interest (see MIL-C-25666)

Preparing activity:
Air Force - 11

(Project 1660 - F361)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE
MIL-C-27336B (USAF) 20 SEP 1978

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

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